

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Research Review & Analysis

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO4: Develop intelligent software agents using suitable agent architectures and learning techniques.(BTL 3)

Assessment Tool Weightage: 30%

Total Marks: 100

Time: 90 Mins

No. of Questions: 1

Annexure A Research Review & Analysis

Code of Conduct:

I _Juhi Surin (192525148)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Juhi Surin

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Domain Understanding	10%	
2. Literature Search and Source Selection	10%	
3. Literature Review and Synthesis	15%	
4. Comparative Analysis of AI Techniques	10%	
5. Critical Analysis of Research Findings	10%	
6. Research Gap Identification	10%	
7. Proposed Research Direction	10%	
8. Research Methodology / Analysis Framework	5%	
9. Experimental / Evidence-Based Validation	10%	
10. Result Analysis and Interpretation	5%	
11. Research Paper Quality and Referencing	10%	
12. Technical Presentation and Research Defense	5%	
Total	100%	

Signature of the Faculty

Total Marks Awarded

ANSWER ANY ONE QUESTION

S.No.	Research Review & Analysis Question	Primary Topic
1	Decision Tree Learning: Review recent research on ID3, C4.5, and CART. Compare their attribute-selection methods, pruning strategies, interpretability, computational complexity, and classification performance. Identify a research gap and suggest a possible improvement.	Decision Trees
2	Neural Networks vs Statistical Learning: Review and compare research on neural-network and statistical-learning approaches for prediction and classification. Analyze their performance, data requirements, generalization, interpretability, computational cost, and suitability for different applications.	Statistical Learning & Neural Networks
3	Kernel Methods: Analyze research on SVM and kernel methods for non-linear classification. Compare linear, polynomial, and RBF kernels and evaluate their advantages and limitations compared with neural-network-based approaches.	Kernel Methods
4	Reinforcement Learning: Review research on Q-Learning, SARSA, and Deep Q-Networks. Compare their learning mechanisms, exploration strategies, convergence, computational requirements, and real-world applications. Identify challenges and research gaps.	Reinforcement Learning
5	Explainable and Generalizable AI: Review research on Inductive Learning and Explanation-Based Learning and analyze how these approaches contribute to generalization and explainability in AI. Identify current limitations and propose future research directions.	Inductive & Explanation-Based Learning

Structure of the Report:

S.No.	Paper Section	Expected Content
1	Title	Specific and concise title related to the selected research question/topic.
2	Abstract	Brief summary of the research problem, review methodology, major findings, research gap, and conclusion.
3	Keywords	4–6 important keywords related to the selected topic.
4	1. Introduction	Introduce the AI topic, background, importance, motivation, and scope of the review.
5	2. Research Problem and Objectives	Clearly state the research problem and specific objectives of the review.
6	3. Literature Review	Review relevant research papers and summarize major approaches, methodologies, findings, and contributions.
7	4. Comparative Analysis	Compare the reviewed approaches based on suitable parameters such as accuracy, complexity, interpretability, computational cost, scalability, and generalization.
8	5. Critical Analysis	Discuss strengths, weaknesses, limitations, assumptions, and practical challenges identified across the reviewed studies.
9	6. Research Gap	Identify unresolved problems, limitations in existing research, and areas requiring further investigation.
10	7. Proposed Research Direction	Suggest possible solutions, improvements, or future research directions based on the identified gap.
11	8. Discussion	Discuss the overall insights obtained from the review and their implications for AI research and applications.
12	9. Conclusion	Summarize the major findings, key insights, research gap, and future direction.
13	References	List all research papers, books, datasets, websites, and other sources using a consistent citation style.

Evaluation Rubrics:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
1. Research Problem Identification and Understanding	10%	9–10% – Clearly identifies the research problem, objectives, scope, and relevance to AI learning approaches.	7–8% – Research problem and objectives are clearly identified with minor omissions.	5–6% – Basic understanding of the research problem with some gaps.	0–4% – Problem is unclear, inappropriate, or poorly understood.
2. Literature Search and Source Selection	10%	9–10% – Uses diverse, highly relevant, recent, and credible research papers, journals, conferences, and technical sources.	7–8% – Uses relevant and credible sources with minor limitations.	5–6% – Uses a limited number of relevant sources.	0–4% – Sources are insufficient, unreliable, or largely irrelevant.
3. Literature Review and Synthesis	15%	13–15% – Provides a comprehensive review and synthesizes findings across multiple studies rather than merely summarizing them.	10–12% – Good review with meaningful comparison of major studies.	7–9% – Basic review with limited synthesis and comparison.	0–6% – Superficial review with little understanding of existing research.

4. Analysis of AI Techniques / Methods	15%	13–15% – Provides technically accurate and in-depth analysis of relevant methods such as decision trees, statistical learning, neural networks, kernel methods, or reinforcement learning.	10–12% – Provides good technical analysis with minor gaps.	7–9% – Provides basic technical description with limited analysis.	0–6% – Methods are poorly understood or inaccurately described.
5. Comparative Analysis of Research Approaches	10%	9–10% – Critically compares approaches based on accuracy, complexity, interpretability, scalability, computational cost, and application suitability.	7–8% – Provides meaningful comparison using several parameters.	5–6% – Basic comparison with limited criteria.	0–4% – Little or no meaningful comparison.
6. Critical Evaluation of Research Findings	10%	9–10% – Critically evaluates strengths, weaknesses, assumptions, experimental methods, and findings of existing studies.	7–8% – Provides good critical evaluation with minor gaps.	5–6% – Some critical observations are provided but analysis is limited.	0–4% – Mostly descriptive with little critical evaluation.

7. Identification of Research Gaps	10%	9–10% – Clearly identifies significant research gaps, limitations, unresolved issues, and opportunities for further investigation.	7–8% – Identifies relevant research gaps with reasonable justification.	5–6% – Identifies basic gaps but provides limited justification.	0–4% – Research gaps are missing or incorrectly identified.
8. Research Insights and Recommendations	5%	5% – Provides insightful conclusions and technically feasible recommendations for future research or applications.	4% – Provides relevant conclusions and recommendations.	2–3% – Basic conclusions with limited recommendations.	0–1% – Conclusions or recommendations are absent or inappropriate.
9. Technical Report and Referencing	10%	9–10% – Report is well structured, technically accurate, professionally written, and uses a consistent citation/reference style.	7–8% – Well-structured report with minor writing or referencing issues.	5–6% – Adequate report but contains several structural or referencing issues.	0–4% – Poorly structured, inadequately referenced, or incomplete report.
10. Originality, Academic Integrity and Use of Sources	5%	5% – Demonstrates original synthesis, proper citation, and strong academic integrity with no inappropriate copying.	4% – Good originality and proper citation with minor issues.	2–3% – Some original analysis but significant dependence on sources.	0–1% – Evidence of substantial copying, poor citation, or academic-integrity concerns.

Comparative Analysis of ID3, C4.5, and CART Decision Tree Learning Algorithms: A Research Review

Abstract

Decision tree learning remains one of the most widely used approaches for classification tasks in artificial intelligence due to its simplicity and interpretability. This review examines three foundational decision tree induction algorithms, ID3, C4.5, and CART, with respect to their attribute-selection methods, pruning strategies, interpretability, computational complexity, and classification performance. The review synthesizes findings from foundational and contemporary literature to highlight the evolution of decision tree methods from entropy-based single-tree induction to statistically robust, noise-tolerant models. A key research gap is identified in the limited adaptability of classical decision tree algorithms to high-dimensional, noisy, and streaming data. Based on this gap, the review proposes a hybrid adaptive pruning and feature-selection framework combining CART's cost-complexity pruning with C4.5's gain-ratio criterion to improve robustness while preserving interpretability.

Keywords

Decision Trees; ID3; C4.5; CART; Attribute Selection; Pruning; Machine Learning; Classification

1. Introduction

Decision trees are a class of supervised learning models that recursively partition a dataset into subsets based on attribute values, producing a tree-like structure of decisions leading to class labels. Their intuitive, rule-based structure makes them one of the most interpretable machine learning models, widely applied in domains such as medical diagnosis, credit scoring, fraud detection, and intelligent agent decision-making. Among the many decision tree induction algorithms proposed since the 1980s, ID3, C4.5, and CART are the most influential, forming the theoretical basis for later ensemble methods such as Random Forests and Gradient Boosted Trees. This review focuses on these three algorithms, examining their design principles and comparing them across key technical dimensions, with the goal of understanding their relative strengths, limitations, and the research opportunities that remain open.

2. Research Problem and Objectives

Despite decades of development, decision tree algorithms continue to exhibit trade-offs between accuracy, interpretability, and robustness to noisy or high-dimensional data. The research problem addressed in this review is: how do the attribute-selection and pruning mechanisms of ID3, C4.5, and CART affect their interpretability, computational efficiency, and classification performance, and what limitations remain unresolved? The specific objectives of this review are to: (i) summarize the theoretical foundations of ID3, C4.5, and

CART; (ii) compare their attribute-selection criteria and pruning strategies; (iii) evaluate their interpretability, computational complexity, and classification performance; (iv) identify unresolved research gaps; and (v) propose a possible improvement direction.

3. Literature Review

Quinlan (1986) introduced ID3, which uses information gain derived from Shannon entropy to select the attribute that best splits the data at each node. While computationally efficient and easy to interpret, ID3 is limited to categorical attributes, cannot handle missing values, and has no built-in pruning mechanism, making it prone to overfitting on noisy datasets.

Quinlan (1993) later proposed C4.5 as a direct successor to ID3. C4.5 replaces information gain with the gain ratio, which normalizes for attributes with many distinct values, reducing bias toward high-cardinality features. It also introduces support for continuous attributes through dynamic threshold splitting, handles missing values probabilistically, and applies pessimistic error-based post-pruning to reduce overfitting, generally improving generalization over ID3.

Breiman et al. (1984) developed CART independently, using the Gini index (or the Twoing criterion) for attribute selection and restricting splits to strictly binary partitions. A distinguishing feature of CART is its cost-complexity pruning, which explicitly balances tree size against misclassification cost using cross-validation. Unlike ID3 and C4.5, CART supports both classification and regression tasks, and its binary-tree structure later became the foundation for ensemble methods such as Random Forests (Breiman, 2001).

More recent surveys (Rokach & Maimon, 2005; Loh, 2011; Charbuty & Abdulazeez, 2021) confirm that while these classical algorithms remain relevant, they are increasingly used as base learners within ensemble frameworks rather than as standalone classifiers for large-scale or high-dimensional problems, due to their sensitivity to data variance and limited scalability.

4. Comparative Analysis

Table 1 summarizes the comparative characteristics of ID3, C4.5, and CART across the key evaluation parameters considered in this review.

Criterion	ID3	C4.5	CART
Attribute Selection	Information Gain (entropy-based)	Gain Ratio (normalized entropy)	Gini Index (or Twoing criterion)
Handles Continuous Data	No (categorical only)	Yes (dynamic thresholding)	Yes (binary splits)
Handles Missing Values	No	Yes (probabilistic weighting)	Yes (surrogate splits)
Tree Structure	Multi-way splits	Multi-way splits	Strictly binary splits
Pruning Strategy	None (prone to overfitting)	Pessimistic post-pruning	Cost-complexity post-pruning

Interpretability	High (simple rules)	High, slightly complex	Moderate (binary structure less intuitive)
Computational Complexity	Low	Moderate to high	Moderate
Typical Output	Classification only	Classification only	Classification and regression
Classification Performance	Baseline, sensitive to noise	Improved generalization	Robust, competitive with ensembles

Overall, ID3 offers the lowest computational overhead but the weakest robustness, C4.5 provides a balanced improvement in generalization and usability at moderate computational cost, and CART offers the most statistically principled pruning approach along with regression capability, at the cost of slightly reduced interpretability due to its strictly binary tree structure.

5. Critical Analysis

A critical examination of the reviewed literature reveals several strengths and weaknesses. ID3's simplicity makes it pedagogically valuable and computationally attractive, but its inability to handle continuous or missing data and its lack of pruning severely limit its practical applicability to real-world datasets. C4.5 addresses many of ID3's shortcomings, but its pessimistic pruning heuristic lacks a strong statistical justification compared to CART's cross-validated cost-complexity pruning, and its gain ratio can occasionally favor unbalanced splits. CART's binary-split restriction improves statistical rigor and enables regression tasks, but can produce deeper, less human-readable trees for problems with naturally multi-valued categorical attributes. Across all three algorithms, a common limitation is instability: small perturbations in training data can produce substantially different tree structures, an issue that motivated the later development of ensemble methods.

6. Research Gap

While ID3, C4.5, and CART are individually well-studied, the literature reveals a persistent research gap in adapting classical single-tree induction methods to modern data characteristics, namely high dimensionality, class imbalance, concept drift in streaming data, and adversarial noise. Existing pruning strategies (pessimistic pruning in C4.5, cost-complexity pruning in CART) were designed for static, moderately sized datasets and do not adequately address dynamic or continuously arriving data. Furthermore, limited work systematically integrates the interpretability advantages of single decision trees with the robustness of ensemble methods without sacrificing transparency, which remains critical for explainable AI applications.

7. Proposed Research Direction

Based on the identified gap, this review proposes a hybrid adaptive decision tree framework that combines C4.5's gain-ratio-based attribute selection with CART's cross-validated cost-complexity pruning, augmented with an incremental update mechanism to handle streaming

data and concept drift. Such a framework could periodically re-evaluate split thresholds using a sliding window of recent instances, allowing the tree to adapt to changing data distributions while retaining a single, interpretable structure. Future work could empirically validate this framework against standard benchmarks and compare its accuracy-interpretability trade-off with ensemble baselines such as Random Forests and XGBoost.

8. Discussion

The comparative and critical analysis presented in this review indicates that no single classical decision tree algorithm is universally superior; rather, the choice depends on the nature of the data and the relative importance of interpretability versus predictive robustness. These findings have broader implications for the design of intelligent software agents, where decision trees are often preferred as interpretable policy or classification components within larger agent architectures. Understanding the trade-offs among ID3, C4.5, and CART allows agent designers to select or hybridize tree-induction strategies appropriate to the agent's operating environment, whether static and well-structured or dynamic and noisy.

9. Conclusion

This review compared ID3, C4.5, and CART decision tree learning algorithms across attribute-selection methods, pruning strategies, interpretability, computational complexity, and classification performance. ID3 provides a simple but limited baseline; C4.5 improves generalization through gain ratio and pruning; and CART contributes statistically robust pruning along with regression capability. The key research gap identified is the limited adaptability of these classical algorithms to high-dimensional and streaming data, and a hybrid adaptive pruning framework was proposed as a promising future research direction that balances interpretability with robustness.

References

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